V01.08 – Mirror Physics II

Mirror Physics II
Constraint Salience and Physical Coupling

Toward Measurable Quantities for Recursive Constraint Structure

Lloyd Christopher Smith

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Abstract

Mirror Physics I identified the physical support conditions required for recursive observer-continuity: ordered causal update, finite propagation, local substrate persistence, metastable memory, thermodynamic throughput and error-corrective repair. The present paper asks whether the organisational structure that makes such continuity possible can be assigned a disciplined mathematical quantity. It introduces constraint salience $\Phi_L^{\Gamma,\tau}$ as a scale-relative and subsystem-relative functional over physical organisation. Constraint salience is not proposed as a new physical field, a modification of relativity or an additional force. It is a candidate bridge quantity intended to measure how strongly a local subsystem preserves viability-relevant structure through causal coupling, memory, error correction, recursive reliability and cost-sensitive repair. The paper specifies minimal requirements for such a quantity: operational measurability, explicit scale and horizon, compatibility with lower-level symmetries, thermodynamic accounting, component decomposability and recovery of established physics. It then proposes a general salience functional, proves that ungoverned complexity is not sufficient for salience, distinguishes weak, intermediate and strong senses of physical coupling, and states the conditions that any future strong-coupling proposal would have to satisfy. The conclusion is conservative: Mirror Theory has not yet produced new physics, but it has isolated the mathematical form of the object that any future Mirror-physical extension would have to define, measure and constrain.

Keywords: Mirror Theory; constraint salience; observerhood; physical coupling; information; thermodynamics; error correction; causal structure; reliability; identity-continuity; recursive systems.

Contents

1 Introduction

Mirror Theory derives observerhood from constraint, viability, bounded modelling, self-modelling, recursive reliability tracking and governed identity-continuity. A local system becomes observer-like not merely because it stores information or predicts its environment, but because self-model reliability becomes action-guiding, cost-sensitive and positively coupled to viability under perturbation. A persistent observer is therefore not a static entity. It is a local process that preserves a governed self-model through time.

Mirror Physics I asked what physical reality must provide for such a process to exist. Its answer was not a new law of nature, but a set of physical support conditions: ordered causal update, local subsystem persistence, finite propagation channels, metastable memory, thermodynamic throughput and error correction. The present paper asks the next question:

> *Can the organisational structure that makes observer-continuity possible be assigned a measurable mathematical quantity?*

This paper calls that quantity *constraint salience*. The notation used here is

$$\Phi_L^{\Gamma,\tau}, \tag{1}$$

where L denotes the local subsystem being evaluated, Γ denotes the chosen coarse-graining or descriptive scale, and τ denotes the temporal horizon over which persistence is assessed.

The notation is intentionally cautious. $\Phi_L^{\Gamma,\tau}$ is not introduced as a pointwise scalar field on spacetime. It is not asserted to couple to gravity, electromagnetism, quantum fields or the metric. It is introduced as a scale-relative functional over organised physical processes. Its purpose is to make explicit what Mirror Theory would have to measure before any future physical extension could be taken seriously.

1.1 Central distinction

The central distinction is this:

> *Complexity is not salience. Information is not salience. Energy flow is not salience. Salience arises only when organised constraint contributes to persistence, correction, self-maintenance and viability-relevant continuity at a specified scale.*

A turbulent fluid may be complex. A random string may have high algorithmic incompressibility. A hot gas may have high microscopic activity. None of these facts alone establishes observer-relevant salience. Mirror salience is not mere richness of structure. It is governed, viability-relevant and persistence-supporting constraint organisation.

1.2 Scope and non-claims

This paper does not claim that $\Phi_L^{\Gamma,\tau}$ modifies the speed of light, changes the Einstein field equations, generates a new force or explains consciousness. It also does not claim that the universe optimises for observers. Its aim is narrower:

> *To define the type of mathematical quantity that would be needed if recursive constraint structure were ever to be made physically explicit.*

That is already a significant step. Without such a quantity, talk of "constraint physics" remains metaphorical. With such a quantity, the programme becomes at least mathematically inspectable. The difference between metaphor and early formalism is not certainty. It is constraint.

2 Position within the Physics Support Arc

The Physics Support Arc begins after the Theory Arc, Mathematical Formalisation Arc and Computational Observerhood Labs. Its function is not to replace established physics. Its function is to ask what established physical regimes must provide for the formal observer-continuity developed in Volume I to be realised, and whether any further physical quantity can be responsibly defined.

Mirror Physics I established a support-condition result. If a local system physically realises observer-continuity, then the surrounding regime must support causal ordering, local persistence, metastable memory, viability-relevant interaction and repair under perturbation. Mirror Physics II does not overturn that result. It asks whether the organisation that satisfies those support conditions can be measured.

V01.06: identity-continuity under governed memory update

↓

V01.07: physical support conditions for observer-continuity

↓

V01.08: constraint salience as a measurable organisational quantity

↓

Future work: thermodynamic cost, empirical calibration and possible coupling tests

Figure 1: The local position of Mirror Physics II within the Volume I sequence. The present paper moves from support conditions to a candidate quantity, without claiming that the quantity is a new field or force.

The paper therefore has a constrained role. It is not a final theory of physical salience. It is a formal bridge. It states the object that a future theory would need to define, measure and discipline before making stronger physical claims.

3 From physical support to constraint salience

Mirror Physics I treated a physical support regime as a tuple

$$\mathcal{P} = (\mathcal{M}, \preceq, g, F, D, E, N), \tag{2}$$

where $\mathcal{M}$ is a spacetime, lattice, graph or effective arena of localisation; $\preceq$ is an ordering relation sufficient for causal or transition ordering; g is a metric or effective causal structure where applicable; F denotes fields and material degrees of freedom; D denotes dynamics; E denotes thermodynamic environment; and N denotes noise or perturbation structure.

A local observer-continuity functional was represented as

$$C_L(t, t+\tau) = \mathrm{Cont}\left(I_L(t), I_L(t+\tau), H_{t:t+\tau}\right), \tag{3}$$

where $I_L(t)$ is the identity-state of the local system and $H_{t:t+\tau}$ is the governed history of memory incorporation, perturbation, repair and update.

The natural next question is whether the physical organisation that supports C_L can be quantified. This is the role of $\Phi_L^{\Gamma,\tau}$.

Definition 1 (Constraint salience)**.** *Let L be a local subsystem embedded in a physical support regime $\mathcal{P}$. Let Γ be a specified coarse-graining and let τ be a temporal horizon. The constraint salience $\Phi_L^{\Gamma,\tau}$ is a dimensionless functional intended to measure the degree to which the physical* *organisation of L, at scale Γ over horizon τ, contributes to viability-relevant persistence, error correction, governed memory and observer-continuity.*

This definition is deliberately incomplete until its components are specified. But it already imposes three restrictions.

First, salience is local-system relative. It is not meaningful to ask for the salience of the universe in the abstract without specifying subsystem, scale, horizon and viability context.

Second, salience is horizon-relative. A structure may be highly salient over milliseconds but irrelevant over years. Another may be slow, durable and salient over evolutionary time.

Third, salience is coarse-graining-relative. At atomic resolution, a brain is a many-body physical system. At cognitive resolution, it is a memory-bearing, self-modelling and error-correcting process. The object of salience is not independent of descriptive scale.

4 Requirements for a physically meaningful salience quantity

A salience quantity cannot be introduced merely because it sounds useful. It must satisfy discipline conditions. This section states the minimal requirements that any candidate $\Phi_L^{\Gamma,\tau}$ must satisfy.

4.1 Operational requirement

A candidate salience quantity must be associated with measurable or computable features of systems. For artificial agents, this may include state variables, memory integrity, calibration, repair cost and task viability. For biological systems, it may include metabolic persistence, homeostatic regulation, predictive control, error correction and survival probability. For physical substrates, it may include stability of macrostates, information retention, noise resistance and energetic throughput.

A purely verbal salience measure is not enough. It must be possible, at least in principle, to estimate its components.

4.2 Scale and horizon requirement

$\Phi_L^{\Gamma,\tau}$ must always specify Γ and τ. Without a scale, salience becomes ambiguous. A whirlpool may preserve macroscopic form while exchanging microscopic particles. A living cell may preserve organisation while replacing molecules. A person may preserve identity while revising memories and goals.

Mirror salience therefore tracks organisation, not material sameness. But organisation itself is scale-relative.

4.3 Symmetry compatibility requirement

If $\Phi_L^{\Gamma,\tau}$ is ever proposed as physically fundamental or physically coupled, it must transform appropriately under the symmetries of the lower-level theory. In relativistic settings, any local density derived from Φ would have to be Lorentz-covariant or arise from a clearly specified foliation or coarse-graining. In gauge theories, it must not depend on gauge artefacts. In thermodynamic settings, it must respect energy accounting and entropy production.

The discipline condition is simple:

> *A future Mirror-physical extension must either show that salience transforms correctly under the relevant physical symmetries, or show that salience is emergent only after coarse-graining and therefore not a fundamental microscopic variable.*

4.4 Non-magic requirement

Constraint salience cannot be allowed to do explanatory magic. If $\Phi_L^{\Gamma,\tau}$ is high, one must be able to say why. Which channels are stable? Which memories are preserved? Which perturbations are corrected? Which costs are paid? Which viability outcomes improve?

A salience quantity that cannot be decomposed into inspectable components is not scientifically useful.

4.5 Recovery requirement

Where a candidate salience measure is applied to domains already described by established physics, it must recover those descriptions in the appropriate limit. If no new physical coupling is asserted, this requirement is modest: salience must be computable from ordinary physical, informational and organisational variables. If strong coupling is asserted, the requirement becomes severe: the proposal must reduce to established relativity, quantum theory and thermodynamics in regimes where those theories are already well tested.

5 A candidate salience functional

A minimal candidate functional can be written as

$$\Phi_L^{\Gamma,\tau} = \mathcal{N}\Big[w_I\mathcal{I}_L^{\Gamma,\tau} + w_C\mathcal{C}_L^{\Gamma,\tau} + w_M\mathcal{M}_L^{\Gamma,\tau} + w_E\mathcal{E}_L^{\Gamma,\tau} + w_R\mathcal{R}_L^{\Gamma,\tau} - w_K K_L^{\Gamma,\tau}\Big], \tag{4}$$

where $\mathcal{N}$ is a normalisation map, $w_i \geq 0$ are context-dependent weights and each component is evaluated over a specified domain.

The components are:

- $\mathcal{I}_L^{\Gamma,\tau}$: viability-relevant information structure;
- $\mathcal{C}_L^{\Gamma,\tau}$: causal constraint coupling and partial closure;
- $\mathcal{M}_L^{\Gamma,\tau}$: metastable memory and identity-continuity;
- $\mathcal{E}_L^{\Gamma,\tau}$: error-correction and recovery capacity;
- $\mathcal{R}_L^{\Gamma,\tau}$: recursive reliability and self-model calibration;
- $K_L^{\Gamma,\tau}$: maintenance, repair, sensing, control and computational cost.

Equation (5) is not a law of physics. It is a scaffold. Its value lies in making explicit the kinds of quantities that must be defined if Mirror Theory is to become operational beyond philosophical description.

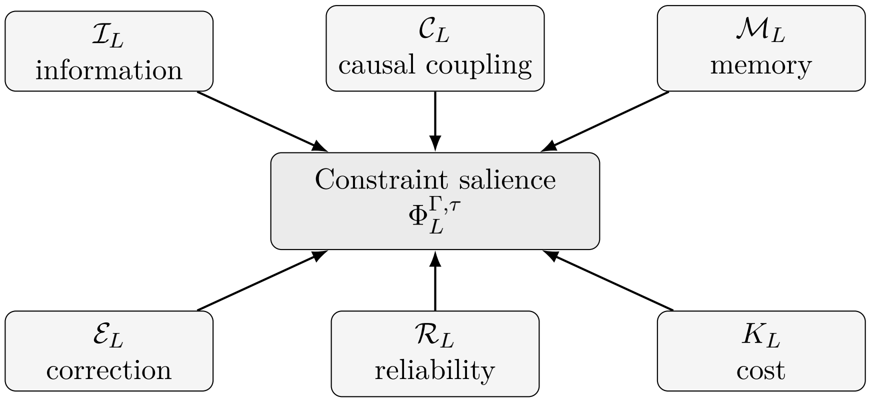

Figure 2: Candidate construction of constraint salience from physical and organisational components. The figure is schematic. No claim is made that these components add linearly in fundamental physics.

6 Components of salience

6.1 Viability-relevant information

The first component is not raw information. It is information relevant to the persistence of the system. A possible form is

$$\mathcal{I}_L^{\Gamma,\tau} = I\big(X_L^\Gamma(t); V_L(t+\tau)\big), \tag{5}$$

where $X_L^\Gamma(t)$ is the coarse-grained state of L and $V_L(t+\tau)$ is future viability. More generally, one may use conditional mutual information:

$$\mathcal{I}_L^{\Gamma,\tau} = I\big(X_L^\Gamma(t); V_L(t+\tau) \mid X_E^\Gamma(t)\big), \tag{6}$$

where X_E denotes environmental state.

This distinguishes useful organisation from arbitrary complexity. Random noise may contain high entropy, but if it does not improve prediction, correction or persistence, it does not generate salience in the Mirror sense.

6.2 Causal constraint coupling

The second component measures whether internal variables constrain one another and the future trajectory of the system. Candidate tools include transfer entropy, causal emergence, effective information, dynamical systems stability and intervention-based causal modelling. A simple abstract representation is

$$\mathcal{C}_L^{\Gamma,\tau} = \mathrm{Coup}\left(X_L^\Gamma(t), X_L^\Gamma(t+\tau), \mathrm{do}(\cdot)\right), \tag{7}$$

where Coup measures the intervention-relevant dependence of future local states on present local organisation.

This term matters because a mere collection of parts is not yet a salient system. A system becomes more salient when its parts mutually constrain future organisation in ways that preserve viability.

6.3 Metastable memory and identity-continuity

The third component measures whether the system physically preserves self-relevant state through time. Using the notation of Mirror Mathematics III,

$$\mathcal{M}_L^{\Gamma,\tau} = \mathrm{Cont}\left(I_L(t), I_L(t+\tau), H_{t:t+\tau}\right), \tag{8}$$

where I_L is the identity-state and H is the governed update history.

This term is central. An observer is not just a momentary information pattern. It is a continuity-preserving process. If a candidate subsystem cannot preserve memory, self-state or update history across the relevant interval, its salience is limited even if its instantaneous complexity is high.

6.4 Error-correction capacity

The fourth component measures resilience under perturbation. A possible abstract definition is

$$\mathcal{E}_L^{\Gamma,\tau} = \mathbb{E}_{\eta\sim N}\left[\mathbf{1}\{X_L(t+\tau) \in B_V\} \mid X_L(t) + \eta\right], \tag{9}$$

where η is a perturbation drawn from the noise distribution N and B_V is the viable basin.

This links Mirror salience to the physics of robustness. Error correction is not optional. In a noisy universe, persistence requires active or passive mechanisms that return the system toward viability-preserving states.

6.5 Recursive reliability

The fifth component measures whether the system tracks the reliability of its own models, and in stronger cases, the reliability of its reliability estimates. A first-order reliability variable may be represented by $R_L^{(1)}$. A recursive reliability structure may be represented by a hierarchy

$$R_L^{(0)}, R_L^{(1)}, R_L^{(2)}, \ldots, R_L^{(n)}, \tag{10}$$

where $R_L^{(0)}$ is the object-level model state, $R_L^{(1)}$ estimates its reliability and $R_L^{(2)}$ estimates the reliability of that estimate.

Prior computational Mirror labs suggest a disciplined pattern: first-order reliability helps when self-state perturbation is present and repair is affordable; recursive reliability helps only when the first-order estimator itself is corrupted. Thus the salience contribution of reliability must be cost-sensitive:

$$\mathcal{R}_L^{\Gamma,\tau} = \max_n \Big(\mathbb{E}[V_L \mid R_L^{(n)}, \pi_n] - \mathbb{E}[V_L \mid \pi_0] - K(R_L^{(n)}, \pi_n)\Big). \tag{11}$$

This prevents recursive reliability from becoming an infinite regress. Higher-order reliability is justified only where it improves viability after cost.

6.6 Maintenance cost

The final component is negative. Organisation costs something. Memory maintenance, repair, sensing, computation, signalling and metabolic support all require resources. A physically meaningful salience measure must subtract these costs:

$$K_L^{\Gamma,\tau} = K_{\text{memory}} + K_{\text{sensing}} + K_{\text{repair}} + K_{\text{computation}} + K_{\text{control}}. \tag{12}$$

This term is essential because not all additional structure is beneficial. A system that over-audits its memories or recursively checks its own reliability beyond need may reduce viability. This was the lesson of the computational labs: reliability is valuable only when actionable and cost-justified.

7 Complexity is not enough

The concept of salience risks being confused with complexity. The following proposition states why that is wrong.

Proposition 1 (Complexity insufficiency)**.** *High informational complexity of a physical state is not sufficient for high constraint salience.*

Proof. Let X be a state with high entropy or high algorithmic incompressibility. High complexity alone does not imply that X contributes to future viability V_L, preserves identity I_L, supports error correction or guides action under perturbation. A random string can be incompressible while having no causal role in the persistence of any local system. Therefore complexity alone is insufficient for salience. Additional structure is required: viability relevance, causal coupling, memory continuity, correction and cost-sensitive reliability. □

Corollary 1 (Noise exclusion)**.** *Random microscopic fluctuation is not Mirror-salient merely because it is physically real or information-rich.*

This is a crucial protection against overextension. Mirror Theory does not imply that every complex physical process is observer-like. It implies that a special subset of organised processes have salience because they preserve governed continuity under perturbation.

8 Relation to existing physical and information-theoretic ideas

Constraint salience sits near several established research traditions but does not collapse into any one of them.

8.1 Shannon information

Shannon information measures uncertainty and channel capacity. It does not by itself measure meaning, viability or observer-continuity. Mirror salience may use Shannon-style quantities, but it filters them through viability and persistence. Information matters only when it contributes to the maintenance and correction of organised structure.

8.2 Thermodynamics and Landauer cost

Memory and computation are physical. Landauer's principle links logically irreversible information erasure to thermodynamic cost. Bennett's work showed how reversible computation reframes the relationship between information and thermodynamics. Mirror salience does not replace these results. It uses them as constraints. Any physically instantiated observer must pay thermodynamic costs for memory, computation and correction.

8.3 Cybernetics and requisite variety

Ashby's law of requisite variety states that effective control requires sufficient internal variety to match environmental variety. Mirror salience extends this cybernetic insight by adding self-model reliability and identity governance. It is not enough to control the environment; the system must also know when its own self-model is unreliable.

8.4 Active inference and expected free energy

Active inference treats perception and action as processes of free-energy minimisation under generative models. Mirror salience overlaps with active inference in its use of prediction, uncertainty and action. The difference is target and threshold. Mirror Theory asks when self-model reliability becomes a viability-relevant condition for observerhood. Constraint salience therefore includes not only uncertainty reduction but persistence of governed identity under perturbation.

8.5 Error correction and quantum information

Modern physics already recognises that information preservation and error correction can be physically deep, especially in quantum information theory. Mirror salience is not a quantum error-correction proposal. However, it treats error correction as a general physical precondition of persistent observerhood. Without mechanisms for preserving state against noise, continuity collapses.

9 Scale, coarse-graining and emergence

One danger is to ask whether Φ exists at a point. That is usually the wrong question. Constraint salience is naturally coarse-grained.

Definition 2 (Coarse-grained salience)**.** *Let $\Gamma : S_{\text{micro}} \to S_{\text{macro}}$ be a coarse-graining map from microphysical states to macrostates. A salience functional $\Phi_L^{\Gamma,\tau}$ is coarse-grained if its value depends on macrostate trajectories*

$$\Gamma(x_t), \Gamma(x_{t+1}), \ldots, \Gamma(x_{t+\tau}) \tag{13}$$

rather than on exact microstate identity.

This is not a weakness. Many physically meaningful quantities are scale-relative. Temperature is not a property of a single molecule in the same way it is a property of a gas. Elasticity is not visible in one atom but appears at material scale. Biological viability is not a property of isolated molecules but of organised networks. Mirror salience is similar: it belongs to organised trajectories at the right scale.

9.1 Equivalence classes of organisation

A person can remain the same observer while exchanging atoms, forgetting details, revising beliefs and healing tissue. This would be impossible if identity required microphysical sameness. Mirror continuity therefore tracks constraint-preserving organisation, not particle identity.

In physical terms, this means salience should be evaluated over equivalence classes of microstates:

$$[x]_\Gamma = \{y \in S_{\text{micro}} : \Gamma(y) = \Gamma(x)\}. \tag{14}$$

The relevant question is whether the trajectory remains within an organisation-preserving equivalence class, not whether every microscopic degree of freedom is unchanged.

Proposition 2 (Scale dependence without arbitrariness)**.** *Scale-relativity does not make salience arbitrary if Γ, τ, L, V_L and the perturbation class N are specified before evaluation.*

Proof. A quantity is arbitrary when its terms can be shifted after the fact to rescue a desired conclusion. The salience functional avoids this when the subsystem, coarse-graining, horizon, viability criterion and perturbation class are fixed in advance. Under those conditions, different evaluators may dispute whether the chosen scale is useful, but they are evaluating the same object. Scale-relativity therefore introduces contextual dependence, not uncontrolled freedom. □

10 Physical coupling: three meanings

The phrase "physical coupling" must be handled carefully. It can mean at least three different things.

10.1 Weak coupling: descriptive dependence

In the weakest sense, $\Phi_L^{\Gamma,\tau}$ is physically coupled because it is computed from physical states. This is uncontroversial. The salience of a biological organism or artificial agent depends on its material organisation. But this does not imply new physics.

10.2 Intermediate coupling: emergent causal relevance

In an intermediate sense, $\Phi_L^{\Gamma,\tau}$ may be causally relevant at an emergent level. For example, memory integrity, metabolic stability and self-model reliability can influence future action. This is physical causation through ordinary mechanisms, described at a higher level. Again, no fundamental law is modified.

10.3 Strong coupling: fundamental or quasi-fundamental influence

The strongest sense would be a claim that salience enters physical equations, modifies causal propagation, affects geometry or changes field dynamics. This paper does not make that claim. But it can state what would be required.

A future strong-coupling proposal would have to specify an action, dynamical equation or conservation relation. A merely illustrative form might be

$$S = S_{\text{phys}} + \epsilon \int_{\Omega} \mathcal{L}_{\Phi}(\Phi, \nabla\Phi, \psi, g)\sqrt{-g}\, d^4x, \tag{15}$$

where ψ denotes physical fields and g the metric. This expression is not proposed as true. It merely shows the level of mathematical discipline required. One would need to define Φ locally or explain its coarse-grained nonlocality, specify dimensions, preserve covariance, avoid superluminal signalling, recover known physics as $\epsilon \to 0$ and derive a testable prediction.

Theorem 1 (No physical coupling without compatibility)**.** *No candidate constraint-salience quantity may be treated as physically coupled to fundamental dynamics unless it satisfies symmetry compatibility, dimensional consistency, recovery of established limiting cases and at least one falsifiable prediction.*

Proof. If the quantity fails symmetry compatibility, it conflicts with the transformation structure of the lower-level physical theory or depends on unphysical representational artefacts. If it fails dimensional consistency, it cannot enter equations in a well-defined way. If it fails to recover established limiting cases, it contradicts well-tested physics without explanation. If it yields no falsifiable prediction, it is not a physical extension but an interpretation or re-description. Therefore all listed conditions are necessary for treating salience as physically coupled to fundamental dynamics. □

11 The speed of light and causal propagation

The original motivation for returning to physics included the invariant speed c. The disciplined position is as follows.

Special relativity treats c as the invariant speed appearing in the structure of spacetime, not merely as the measured speed of visible light. Mirror Physics I interpreted c as part of the causal support architecture that makes stable observer-continuity possible. The present paper does not alter that claim.

A Mirror framing may say:

> *A universe containing persistent recursive observers must support stable causal ordering. In our universe, relativistic causal structure supplies such ordering through light-cone structure and the invariant speed c.*

It may not yet say:

> *Constraint salience changes c.*

That would require a strong-coupling theory, and no such theory has been derived here.

11.1 Why not introduce an effective speed?

Earlier speculative discussions considered a toy expression of the form

$$c_{\text{eff}} = c(1 + \epsilon\Phi). \tag{16}$$

This paper deliberately avoids using that as a physical proposal. The reason is simple: without an action principle, symmetry analysis, limiting-case recovery and empirical prediction, such an equation would be arbitrary.

It may still be useful as a heuristic in private exploration, but it should not be presented as a result. The mature move is to define $\Phi_L^{\Gamma,\tau}$ first, identify its possible measurements and only later ask whether any coupling to causal propagation is mathematically defensible.

Proposition 3 (No propagation modification from salience alone)**.** *The definition of $\Phi_L^{\Gamma,\tau}$ does not, by itself, imply any modification of relativistic causal propagation.*

Proof. $\Phi_L^{\Gamma,\tau}$ is defined as an organisational functional over a local subsystem at a chosen scale and horizon. Its components measure information relevance, causal coupling, memory, error correction, reliability and cost. None of these definitions introduces a new dynamical equation for spacetime, a modified metric, a new field equation or a mechanism for superluminal signalling. Therefore the salience definition alone has no implication for propagation speed. Any such implication would require an additional coupling theory, which this paper does not supply. □

12 A salience threshold for physical observerhood

The salience functional allows a physical version of the minimal observer threshold.

Definition 3 (Physical observer-salience threshold)**.** *A local subsystem L crosses a physical observer-salience threshold at scale Γ and horizon τ if*

$$\Phi_L^{\Gamma,\tau} > \theta_\Phi \tag{17}$$

and if the positive contribution of the salience components to observer-continuity exceeds maintenance cost under the relevant perturbation distribution.

This definition is intentionally not equivalent to consciousness. A system may cross a minimal observer-salience threshold without phenomenal experience. The threshold concerns physical support for recursive observer-continuity, not subjective feel.

Proposition 4 (Salience implies organised persistence). *If $\Phi_L^{\Gamma,\tau} > \theta_\Phi$ under a properly normalised functional whose components satisfy the operational requirements above, then L exhibits non-trivial organised persistence at scale Γ over horizon τ.*

Proof. By construction, $\Phi_L^{\Gamma,\tau}$ exceeds threshold only when its positive components - viability-relevant information, causal coupling, memory continuity, error correction and recursive reliability - jointly outweigh maintenance costs. Each positive component contributes to persistence of organised structure over the horizon τ. Therefore threshold-crossing entails non-trivial organised persistence, provided the normalisation and component definitions are non-degenerate. □

13 Boundary cases

The usefulness of $\Phi_L^{\Gamma,\tau}$ depends on whether it can distinguish cases that ordinary complexity language blurs. Table 1 is qualitative and diagnostic. It is not a metaphysical ranking.

System	**Likely salience**	**Reason**	**Mirror interpretation**
Random gas	Low	High microactivity but little governed identity or correction.	Not observer-salient.
Crystal	Low to moderate	Stable structure but limited self-maintenance or adaptive repair.	Constraint-stable, not observer-like.
Thermostat	Low	Feedback control without robust self-model reliability.	Cybernetic but below observer threshold.
Cell	Moderate	Metabolism, repair, boundary maintenance and viability.	Primitive organisational salience.
Animal nervous system	High	Integrated modelling, regulation, memory and adaptive correction.	Strong biological salience.
Stateless LLM call	Low to moderate	Rich semantic output without durable governed identity.	Not persistent observerhood.
Memory-augmented agent	Moderate	Persistence possible but vulnerable to contamination.	Requires governance.
Mirror-style agent	Higher in relevant regimes	Commit gates, self-model reliability and repair policies.	Candidate artificial observer-salience.
Human person	High	Embodied memory, self-models, repair, continuity and social correction.	Strong observer-continuity.

Table 1: Boundary cases for constraint salience. Categories are qualitative and depend on scale, horizon and component definitions.

The question is not which systems are important. The question is which systems instantiate governed, viability-relevant constraint persistence.

14 Testability and measurement

At present, $\Phi_L^{\Gamma,\tau}$ is most testable in artificial systems, because the relevant variables can be instrumented. One can measure memory contamination, repair cost, self-model error, reliability calibration, task success and viability. The Mirror Observerhood Lab sequence was designed for this reason.

For biological systems, measurement is harder but not impossible. Candidate proxies include homeostatic stability, allostatic regulation, metabolic repair, sensorimotor calibration, nervous system error correction, memory reliability and behavioural recovery after perturbation.

For fundamental physics, no direct salience measurement is currently proposed. The correct discipline is:

1. Define salience operationally in artificial and biological systems.
2. Determine whether the components correspond to known physical quantities: free energy, information flow, error correction, memory stability and entropy production.
3. Ask whether any invariant or approximately conserved structure appears across scales.
4. Only then investigate whether a deeper physical coupling is meaningful.

14.1 Possible empirical programmes

Artificial agents. Build agents with different combinations of memory, self-models, reliability variables and repair policies. Measure $\Phi_L^{\Gamma,\tau}$ proxies and compare them to viability under perturbation.

Biological regulation. Study systems that maintain identity under material turnover: cells, immune systems, nervous systems and developmental systems. Ask how memory, repair and boundary maintenance contribute to persistence.

Physical substrates. Investigate how metastable memory, error correction and thermodynamic cost constrain the possible realisation of persistent observers in physical media.

15 Limitations

This paper is deliberately preliminary. Its limitations are substantial.

First, $\Phi_L^{\Gamma,\tau}$ is not unique. Different normalisations and component choices may be appropriate for different systems. This is not fatal, but it means the framework requires calibration.

Second, the functional in Equation (5) is not derived from fundamental physics. It is a candidate organisational measure. It becomes physically deeper only if future work connects it to lower-level dynamics.

Third, the weights w_i are context-sensitive. A mature theory would need principled methods for assigning them.

Fourth, the framework risks circularity if viability, identity and salience are defined in terms of one another without independent operational anchors. This is why artificial-agent experiments are important: they provide explicit state variables and measurable outcomes.

Fifth, nothing here modifies relativity, quantum mechanics or thermodynamics. Any future attempt to do so would require far more than the present formalism.

16 Roadmap to Mirror Physics III

The next safe step is not a claim about faster-than-light travel or spacetime engineering. The next step is thermodynamic.

A natural sequel would be:

> *Mirror Physics III: Thermodynamic Observerhood, Memory, Error Correction and the Energetic Cost of Persistent Identity.*

Such a paper would examine how memory, repair, computation and identity-continuity are physically paid for. It would connect Mirror salience to Landauer cost, entropy production, homeostasis, free energy and error correction.

Only after that would it be responsible to revisit stronger questions about causal propagation, invariant speeds or possible coupling to spacetime geometry.

17 Conclusion

Mirror Physics II introduced constraint salience $\Phi_L^{\Gamma,\tau}$ as a candidate organisational quantity for recursive constraint structure. The proposal is intentionally conservative. $\Phi_L^{\Gamma,\tau}$ is not a new force, not a new field and not a modification of relativity. It is a scale-relative functional intended to measure the degree to which local physical organisation supports viability-relevant persistence, memory governance, error correction, recursive reliability and observer-continuity.

The paper's main contribution is not a new physical law. It is a discipline condition. If Mirror Theory is ever to matter physically, it must define a measurable salience quantity, specify scale and horizon, respect lower-level symmetries, account for thermodynamic cost, recover established physics and produce testable consequences before proposing deviations.

The central result can be stated plainly:

> *Constraint salience is the bridge quantity Mirror Theory would need before it could responsibly move from observerhood mathematics to physical coupling.*

That bridge is now defined, but not crossed. The next work must test, refine and physically ground it.

A Compact formal summary

A physical support regime:

$$\mathcal{P} = (\mathcal{M}, \preceq, g, F, D, E, N). \tag{18}$$

A local observer-continuity functional:

$$C_L(t, t+\tau) = \mathrm{Cont}\left(I_L(t), I_L(t+\tau), H_{t:t+\tau}\right). \tag{19}$$

A candidate salience functional:

$$\Phi_L^{\Gamma,\tau} = \mathcal{N}\Big[w_I \mathcal{I}_L^{\Gamma,\tau} + w_C \mathcal{C}_L^{\Gamma,\tau} + w_M \mathcal{M}_L^{\Gamma,\tau} + w_E \mathcal{E}_L^{\Gamma,\tau} + w_R \mathcal{R}_L^{\Gamma,\tau} - w_K K_L^{\Gamma,\tau}\Big]. \tag{20}$$

A physical observer-salience threshold:

$$\Phi_L^{\Gamma,\tau} > \theta_\Phi. \tag{21}$$

A strong-coupling discipline condition:

$$\text{coupling claim} \Rightarrow \text{symmetry} + \text{dimensions} + \text{recovery} + \text{prediction}. \tag{22}$$

B Glossary

Constraint salience
A scale-relative quantity measuring the extent to which organised constraint supports viability-relevant persistence.

Coarse-graining A map from lower-level physical states to higher-level macrostates or organisational variables.

Physical coupling A relation between salience and physical dynamics; weak, intermediate and strong forms must be distinguished.

Weak coupling Descriptive dependence of salience on physical states.

Intermediate coupling Emergent causal relevance through ordinary physical mechanisms.

Strong coupling A proposed influence of salience on fundamental or quasi-fundamental dynamics.

Observer-salience threshold A threshold above which a local subsystem exhibits non-trivial organised persistence relevant to observer-continuity.

Non-magic requirement The requirement that a high salience value be decomposable into inspectable channels, memory, repair, cost and viability contributions.

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